



UNIVERSITY OF LEEDS

Formal Language & Finite Automata

Undecidability

Dr Noleen Koehler

University of Leeds

All reading is from the "Introduction to the theory of computation" - Micheal Sipser

- Turing machines - Pages 206 - 2014 (essential)

Existence of unrecognizable languages

Existence of unrecognizable languages

We can prove that there are languages which are not Turing recognizable by a counting argument.

Theorem

There exists a language which is not Turing recognizable.

For the proof we use the following two Lemmas.

Lemma

The set of all strings over an alphabet Σ is countable.

Lemma

The set of all languages over an alphabet Σ is uncountable.

An undecidable and an unrecognizable language

An undecidable and an unrecognizable language

We define the following language:

$$A_{\text{TM}} = \{\langle A, w \rangle \mid A \text{ is a TM that accepts } w\}.$$

Theorem

A_{TM} is not Turing decidable.

Theorem

$\overline{A_{\text{TM}}}$ is not Turing recognizable.